

Lab Experiments 1-2

Data Handling & Visualization: Python + R, Code & Output

Each experiment: Python code + output, followed by R code + output

Date: 15/07/2026

Experiment 1

Monthly Sales Data

PYTHON

Code

```
import matplotlib.pyplot as plt
import numpy as np

# ---- Given dataset: Monthly Sales ----
months = ['January', 'February', 'March', 'April', 'May']
sales = [15000, 18000, 22000, 20000, 23000]

# ---- Task 2 needs "top-selling products" - not in the given table,
#       so a small supplementary product dataset is assumed here ----
products = ['Product A', 'Product B', 'Product C', 'Product D', 'Product E']
product_sales = [42000, 35000, 51000, 28000, 39000]

# ---- Task 3 needs "advertising budget" - not in the given table,
#       so a plausible monthly budget series is assumed here ----
ad_budget = [2000, 2500, 3200, 2800, 3500]

# 1. Line chart - monthly sales trend
plt.figure(figsize=(7, 5))
plt.plot(months, sales, marker='o', color='steelblue')
plt.title("Monthly Sales Trend")
plt.xlabel("Month")
plt.ylabel("Sales (in $)")
plt.tight_layout()
plt.show()

# 2. Bar chart - top-selling products for the year
plt.figure(figsize=(7, 5))
plt.bar(products, product_sales, color='darkorange')
plt.title("Top-Selling Products for the Year")
plt.xlabel("Product")
plt.ylabel("Annual Sales (in $)")
plt.tight_layout()
plt.show()

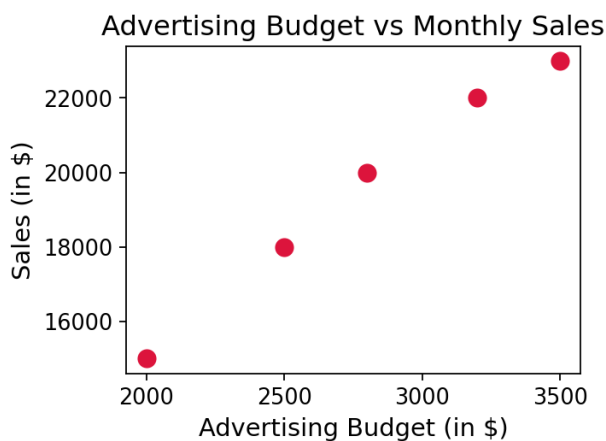
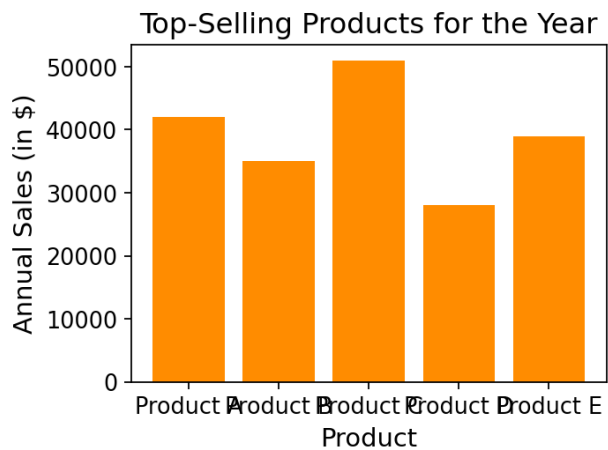
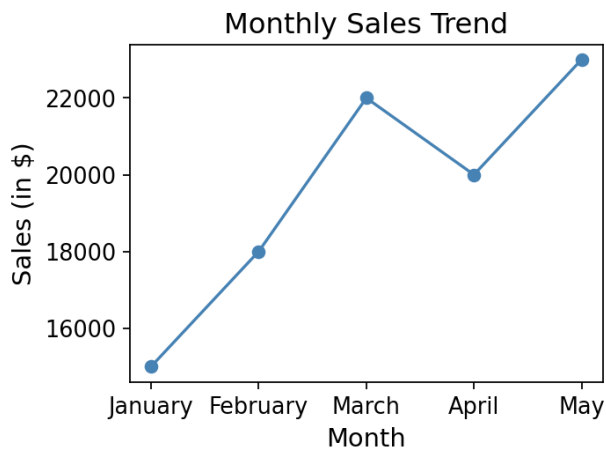
# 3. Scatter plot - advertising budget vs monthly sales
plt.figure(figsize=(7, 5))
plt.scatter(ad_budget, sales, color='crimson', s=80)
plt.title("Advertising Budget vs Monthly Sales")
plt.xlabel("Advertising Budget (in $)")
plt.ylabel("Sales (in $)")
plt.tight_layout()
plt.show()
# Insight: sales rise as ad budget rises, suggesting a positive relationship
# between advertising spend and revenue (correlation, not proof of causation).

# 4. Combined dashboard - line chart + bar chart together
fig, axes = plt.subplots(1, 2, figsize=(13, 5))
axes[0].plot(months, sales, marker='o', color='steelblue')
axes[0].set_title("Monthly Sales Trend")
axes[0].set_xlabel("Month"); axes[0].set_ylabel("Sales (in $)")

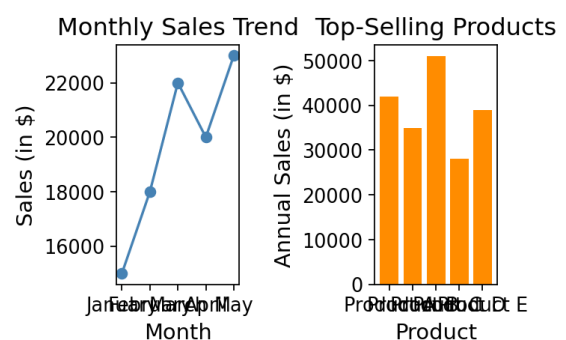
axes[1].bar(products, product_sales, color='darkorange')
axes[1].set_title("Top-Selling Products")
axes[1].set_xlabel("Product"); axes[1].set_ylabel("Annual Sales (in $)")

plt.suptitle("Sales Dashboard: Monthly Trend + Top Products")
plt.tight_layout()
plt.show()
```

Output



Sales Dashboard: Monthly Trend + Top Products



R

Code

```
library(ggplot2)
library(gridExtra)

# ---- Given dataset: Monthly Sales ----
months <- c('January', 'February', 'March', 'April', 'May')
sales <- c(15000, 18000, 22000, 20000, 23000)
df_sales <- data.frame(Month=factor(months, levels=months), Sales=sales)

# ---- Task 2 needs "top-selling products" - not in the given table,
#       so a small supplementary product dataset is assumed here ----
products <- c('Product A', 'Product B', 'Product C', 'Product D', 'Product E')
product_sales <- c(42000, 35000, 51000, 28000, 39000)
df_products <- data.frame(Product=factor(products, levels=products), Sales=product_sales)

# ---- Task 3 needs "advertising budget" - not in the given table,
#       so a plausible monthly budget series is assumed here ----
ad_budget <- c(2000, 2500, 3200, 2800, 3500)
df_ad <- data.frame(Budget=ad_budget, Sales=sales)

# 1. Line chart - monthly sales trend
p1 <- ggplot(df_sales, aes(x=Month, y=Sales, group=1)) +
  geom_line(color="steelblue", size=1) + geom_point(color="steelblue", size=2) +
  labs(title="Monthly Sales Trend", x="Month", y="Sales (in $)") +
  theme_minimal()
print(p1)

# 2. Bar chart - top-selling products for the year
p2 <- ggplot(df_products, aes(x=Product, y=Sales)) +
  geom_col(fill="darkorange") +
  labs(title="Top-Selling Products for the Year", x="Product", y="Annual Sales (in $)") +
  theme_minimal()
print(p2)

# 3. Scatter plot - advertising budget vs monthly sales
p3 <- ggplot(df_ad, aes(x=Budget, y=Sales)) +
  geom_point(color="firebrick", size=3) +
```

```

labs(title="Advertising Budget vs Monthly Sales", x="Advertising Budget (in $)", y="Sales (in $)") +
theme_minimal()
print(p3)
# Insight: sales rise as ad budget rises, suggesting a positive relationship
# between advertising spend and revenue (correlation, not proof of causation).

# 4. Combined dashboard - line chart + bar chart together
grid.arrange(p1, p2, ncol=2, top="Sales Dashboard: Monthly Trend + Top Products")

```

Output



Experiment 2

Customer Feedback Analysis

PYTHON

Code

```
import matplotlib.pyplot as plt
import pandas as pd
import numpy as np
from wordcloud import WordCloud

# ---- Given dataset: Customer Satisfaction ----
df = pd.DataFrame({
    'CustomerID': [1, 2, 3, 4, 5],
    'Age': [25, 30, 35, 28, 40],
    'SatisfactionScore': [4, 5, 3, 4, 5]
})

# ---- Task 4 needs open-ended feedback text - not in the given table,
# so sample feedback strings are assumed here ----
feedback = [
    "great service fast delivery friendly staff",
    "product quality excellent will buy again",
    "delivery was late but support was helpful",
    "amazing experience highly recommend this store",
    "price a bit high but service is great"
]

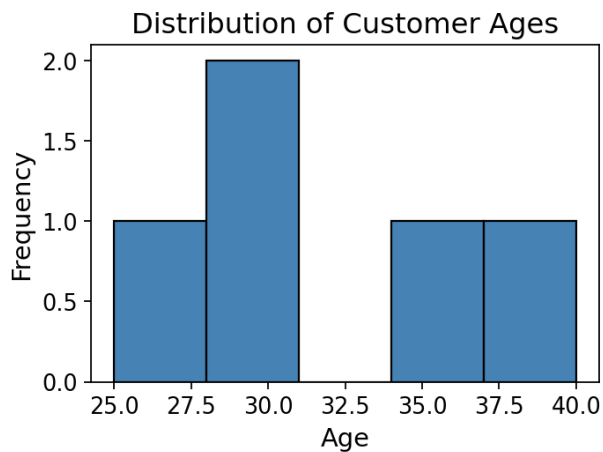
# 1. Histogram - distribution of customer ages
plt.figure(figsize=(7, 5))
plt.hist(df['Age'], bins=5, color='steelblue', edgecolor='black')
plt.title("Distribution of Customer Ages")
plt.xlabel("Age")
plt.ylabel("Frequency")
plt.tight_layout()
plt.show()

# 2. Pie chart - distribution of satisfaction scores
score_counts = df['SatisfactionScore'].value_counts().sort_index()
plt.figure(figsize=(6, 6))
plt.pie(score_counts.values, labels=[f"Score {s}" for s in score_counts.index],
        autopct='%1.1f%%', startangle=90)
plt.title("Distribution of Customer Satisfaction Scores")
plt.tight_layout()
plt.show()

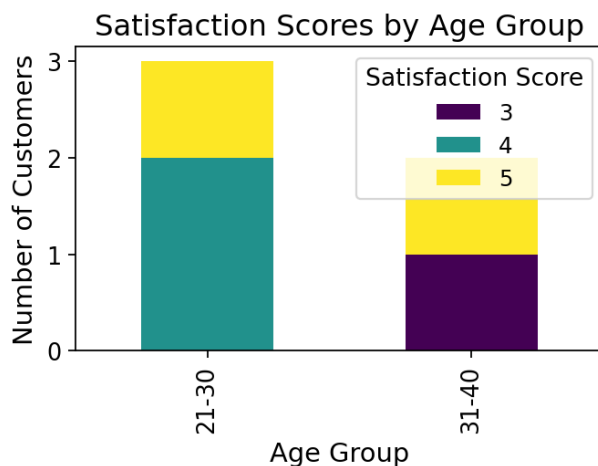
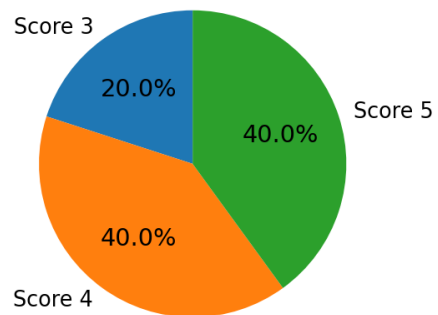
# 3. Stacked bar chart - satisfaction scores by age group
df['AgeGroup'] = pd.cut(df['Age'], bins=[20, 30, 40, 50], labels=['21-30', '31-40', '41-50'])
stacked = df.groupby(['AgeGroup', 'SatisfactionScore'], observed=True).size().unstack(fill_value=0)
stacked.plot(kind='bar', stacked=True, figsize=(7, 5), colormap='viridis')
plt.title("Satisfaction Scores by Age Group")
plt.xlabel("Age Group")
plt.ylabel("Number of Customers")
plt.legend(title="Satisfaction Score")
plt.tight_layout()
plt.show()

# 4. Word cloud - prevalent sentiments from open-ended feedback
text = " ".join(feedback)
wc = WordCloud(width=800, height=400, background_color='white').generate(text)
plt.figure(figsize=(9, 5))
plt.imshow(wc, interpolation='bilinear')
plt.axis('off')
plt.title("Customer Feedback Word Cloud")
plt.tight_layout()
plt.show()
```

Output



Distribution of Customer Satisfaction Scores



R

Code

```
library(ggplot2)
library(wordcloud)
library(tm)
library(dplyr)

# ---- Given dataset: Customer Satisfaction ----
df <- data.frame(
  CustomerID = 1:5,
  Age = c(25, 30, 35, 28, 40),
  SatisfactionScore = c(4, 5, 3, 4, 5)
)

# ---- Task 4 needs open-ended feedback text - not in the given table,
# so sample feedback strings are assumed here ----
feedback <- c(
  "great service fast delivery friendly staff",
  "product quality excellent will buy again",
  "delivery was late but support was helpful",
  "amazing experience highly recommend this store",
  "price a bit high but service is great"
)

# 1. Histogram - distribution of customer ages
hist(df$Age, breaks=5, col="steelblue", border="black",
     main="Distribution of Customer Ages", xlab="Age", ylab="Frequency")

# 2. Pie chart - distribution of satisfaction scores
score_counts <- table(df$SatisfactionScore)
pie(score_counts, labels=paste0("Score ", names(score_counts)),
     main="Distribution of Customer Satisfaction Scores",
     col=rainbow(length(score_counts)))

# 3. Stacked bar chart - satisfaction scores by age group
df$AgeGroup <- cut(df$Age, breaks=c(20,30,40,50), labels=c('21-30', '31-40', '41-50'))
stacked <- df %>% group_by(AgeGroup, SatisfactionScore) %>% summarise(Count=n(), .groups='drop')
```

```

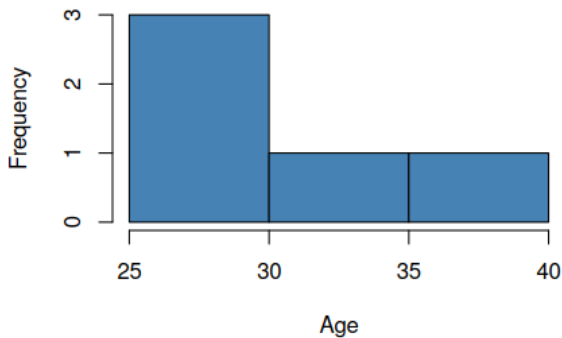
p3 <- ggplot(stacked, aes(x=AgeGroup, y=Count, fill=factor(SatisfactionScore))) +
  geom_col() +
  labs(title="Satisfaction Scores by Age Group", x="Age Group", y="Number of Customers",
    fill="Satisfaction Score") +
  theme_minimal()
print(p3)

# 4. Word cloud - prevalent sentiments from open-ended feedback
corpus <- Corpus(VectorSource(feedback))
corpus <- tm_map(corpus, content_transformer(tolower))
tdm <- TermDocumentMatrix(corpus)
word_freq <- sort(rowSums(as.matrix(tdm)), decreasing=TRUE)
wordcloud(names(word_freq), word_freq, min.freq=1, colors=brewer.pal(8, "Dark2"))
title("Customer Feedback Word Cloud")

```

Output

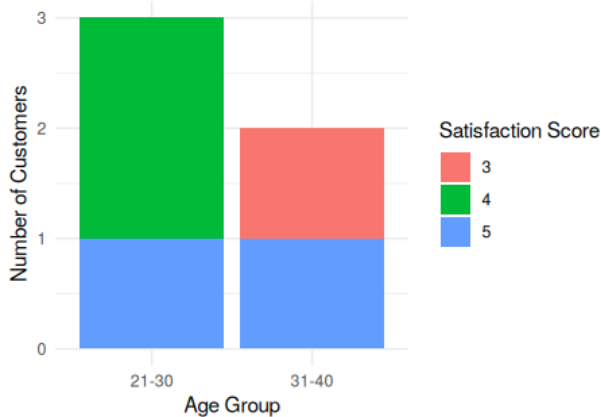
Distribution of Customer Ages



Distribution of Customer Satisfaction Scores



Satisfaction Scores by Age Group



Customer Feedback Word Cloud

